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INTERESTS                • Microlocal Sheaf Theory  
                               • D-modules  
                               • Geometric Representation Theory  
                               • Symplectic and Contact Geometry  
                               • Noncommutative geometry  
                               • Mirror symmetry

EDUCATION AND TRAINING    • Ph.D. in Mathematics, UC Berkeley, 2016 - 2022, awarded on May 13, 2022.  
                                     Advisor: Vivek Shende.  
                                     Thesis title: Symplectic geometric methods in microlocal sheaf theory.  
                               • B.S. in Mathematics, National Taiwan University, Taipei, Taiwan, 2010 - 2014.  
                                     Advisor: Chung-Jun Tsai.  
                                     Thesis title: A calculation for the Hard Lefschetz map on six-manifolds constructed by Coisotropic Luttinger Surgery.

APPOINTMENTS           • Postdoc at the Max Planck Institute for Mathematics, Summer 2025 - Present.  
                               • Postdoc for Noncommutative Algebraic Geometry, SLMath, Spring 2024.  
                               • Assistant Professor (RTPC) of Mathematics, University of Southern California, Fall 2022 - Spring 2025.  
                                     — Herbert and Ruth Busemann Assistant Professor of Mathematics, Fall 2023 - Spring 2024.

RESEARCH                • Tomohiro Asano, Yuichi Ike, C. Kuo, and Wenyan Li. *Sheaf quantization of non-exact Lagrangians and applications*, in preparation.  
                               • Sheel Ganatra, C. Kuo, Wenyan Li, and Haosen Wu. *A microlocal criterion for compactness preserving integral kernels*, in preparation.  
                               • Tomohiro Asano, Yuichi Ike, C. Kuo, and Wenyan Li.  *$C^0$ -rigidity of Legendrians and coisotropics via sheaf quantization*, [arXiv:2510.01746](https://arxiv.org/abs/2510.01746), submitted in October 2025.  
                               • C. Kuo and Harold Williams. *Tropical Lagrangian coamoebae and free resolutions*. [arXiv:2412.18071](https://arxiv.org/abs/2412.18071).  
                               • C. Kuo and Wenyan Li. *Relative Calabi-Yau structure on microlocalization*. [arXiv:2408.04085](https://arxiv.org/abs/2408.04085), submitted in August 2025.  
                               • C. Kuo and Wenyan Li. *Duality and Kernels in Microlocal Geometry*, IMRN 2025 (2025): [rnaf070](https://arxiv.org/abs/2405.15211). ([arXiv:2405.15211](https://arxiv.org/abs/2405.15211))  
                               • Laurent Côté, C. Kuo, David Nadler, and Vivek Shende. *The microlocal Riemann-Hilbert correspondence for complex contact manifolds*. [arXiv:2406.16222](https://arxiv.org/abs/2406.16222), submitted in April 2025.  
                               • C. Kuo, Vivek Shende, and Bingyu Zhang. *On the Hochschild cohomology of Tamarkin categories*. [arXiv:2312.11447](https://arxiv.org/abs/2312.11447), submitted in June 2025.  
                               • C. Kuo and Wenyan Li. *Spherical adjunction and Serre functor from microlocalization*. [arXiv:2210.06643](https://arxiv.org/abs/2210.06643), submitted in February 2025.

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|                           | <ul style="list-style-type: none"> <li>• Laurent Côté, C. Kuo, David Nadler, and Vivek Shende. <i>Perverse Microsheaves</i>. <a href="https://arxiv.org/abs/2209.12998">arXiv:2209.12998</a>, accepted by Compositio.</li> <li>• C. Kuo. <i>Wrapped sheaves</i>, <i>Advances in Mathematics</i> 415 (2023): 108882. (<a href="https://arxiv.org/abs/2102.06791">arXiv:2102.06791</a>)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| PH.D. STUDENTS SUPERVISED | <ul style="list-style-type: none"> <li>• Haosen Wu, USC, co-supervised with Wenyuan Li under Sheel Ganatra.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| TEACHING EXPERIENCE       | <ul style="list-style-type: none"> <li>• Instructor, V5D4, Introduction to Microlocal Sheaf Theory, University of Bonn, Summer 2026, notes available at <a href="https://chriskuo0.github.io/homepage/V5D4-ItMST/">https://chriskuo0.github.io/homepage/V5D4-ItMST/</a></li> <li>• Instructor, Math 125, Calculus I, University of Southern California, Spring 2025</li> <li>• Instructor, Math 425a, Fundamental Concepts of Analysis, University of Southern California, Spring 2025</li> <li>• Instructor, Math 425a, Fundamental Concepts of Analysis, University of Southern California, Fall 2024</li> <li>• Instructor, Math 144, Foundations of Statistics, University of Southern California, Fall 2023</li> <li>• Instructor, Math 245, Mathematics of Physics &amp; Engineering I, University of Southern California, Spring 2023</li> <li>• Instructor, Math 125g, Calculus I, University of Southern California, Fall 2022</li> <li>• Graduate Student Instructor, Math 110, Linear Algebra, UC Berkeley, Spring 2020</li> <li>• Graduate Student Instructor, Math 16B, Analytic Geometry and Calculus, UC Berkeley, Spring 2019</li> <li>• Graduate Student Instructor, Math 54, Linear Algebra and Differential Equations, UC Berkeley, Fall 2017 and Fall 2018</li> <li>• Graduate Student Instructor, Math 53, Multivariable Calculus, UC Berkeley, Spring 2017</li> <li>• Graduate Student Instructor, Math 1A, Calculus, UC Berkeley, Fall 2016</li> <li>• Teaching Assistant, Geometry (Required course for juniors), National Taiwan University, Fall 2013</li> </ul> |
| SEMINARS                  | <ul style="list-style-type: none"> <li>• Organizer, Bonn Symplectic Geometry Seminar, University of Bonn, Fall 2025 - Present</li> <li>• Organizer, Microlocal learning seminar, University of Southern California, Spring 2025</li> <li>• Co-organizer, Topology seminar, University of Southern California, Fall 2022 - Spring 2025</li> <li>• Organizer, Deformation quantization modules learning seminar, Syddansk Universitet, Fall 2021 - Spring 2022</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| INVITED TALKS             | <ul style="list-style-type: none"> <li>• Characteristic cycles on contact manifolds, Topology Seminar, USC, August 2026.</li> <li>• Perverse microsheaves on contact manifolds, seminar at the Chair of Arithmetic Geometry, EPFL, November 2025.</li> <li>• Riemann-Hilbert on contact manifolds, MPI-Oberseminar, MPIM, August</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |

2025.

- Tropical Lagrangian coamoebae and free resolutions, Syzygies and Mirror Symmetry Virtual Seminar, April 2025.
- Symplectic Geometry and Sheaves, Geometry and Topology Seminar, UC Irvine, November 2024.
- A quick introduction of microlocal sheaf theory, NAG GRT Seminar, SLMath, February 2024.
- Globalizing the microlocal Riemann-Hilbert correspondence, Algebraic Geometry Seminar, UC Davis, January 2024.
- Globalizing the microlocal Riemann-Hilbert correspondence, Algebraic Geometry Seminar, Academia Sinica, January 2024.
- Globalizing the microlocal Riemann-Hilbert correspondence, Geometry Seminar, Kyushu University, January 2024.
- Microlocal Riemann-Hilbert correspondence, QM Research Seminar, University of Southern Denmark, December 2023.
- Microlocal Riemann-Hilbert correspondence, Bonn symplectic geometry seminar, December 2023.
- Perverse Microsheaves, Mathematics Seminars, Montana State University, October 2023.
- Perverse Microsheaves, Differential geometry and symplectic topology seminar, University of Minnesota, September 2023.
- Sheaf Theoretic Methods in Symplectic Geometry, NCTS Differential Geometry Seminar, National Center for Theoretical Sciences, Taipei City, May 2023.
- Microlocal sheaf theory in noncommutative geometry, Algebraic Geometry Seminar, Academia Sinica, Taipei City, May 2023.
- Microlocal sheaf theory in noncommutative geometry, UCSB Seminar on Geometry and Arithmetic, University of California, Santa Barbara, May 2023.
- Wrapped sheaves, Topology Seminar in USC, September 2022
- Wrapped sheaves, QM Research Seminar in SDU, March 2022
- Wrapped sheaves, U. Iowa Topology Seminar, December 2021

#### SEMINAR AND TALKS

- Recollections on analytic stacks, Arbeitsgruppe Arithmetische Geometrie und Darstellungstheorie (ARGOS), MPIM, Fall 2025.
- Sheaf quantization and the topology of cotangent bundles, guest lecture in Sheel Ganatra's Math 699, USC, Fall 2022
- Nondisplaceability results for Lagrangians in toric symplectic manifolds, Southern California Symplectothon, October 2022
- Hochschild homology, Deformation quantization modules learning seminar, Syddansk Universitet, February 2022
- Kernels, Deformation quantization modules learning seminar, Syddansk Universitet, December 2021
- Algebroids, Deformation quantization modules learning seminar, Syddansk Universitet, November 2021
- Nonlinear regularity and local existence, J-Holomorphic curves learning seminar, Syddansk Universitet, October 2021
- Deformation of a sheaf of rings, Deformation quantization modules learning

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|                      | seminar, Syddansk Universitet, October 2021                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|                      | <ul style="list-style-type: none"> <li>• <math>E_n</math>-algebras and iterated loop spaces, Geometric Representation Theory seminar, UC Berkeley, Spring 2020</li> <li>• Speaker, Wrapped Floer Homology Learning Seminar, Fall 2019</li> <li>• Hochschild-Kostant-Rosenberg theorem through loop spaces, Geometric Representation Theory seminar, UC Berkeley, November 27 and December 14, 2018</li> <li>• Koszul duality for algebras, Deformation Theory Seminar, UC Berkeley, October 15 2017</li> <li>• Gerbes. Mayer-Vietoris calculations, K-theory Seminar, UC Berkeley, April 2017</li> <li>• Adams operation, K-theory Seminar, UC Berkeley, February 2017</li> <li>• Winter Camp for Fundamentals in Mathematics for College Students, National Taiwan University, January 23-February 8, 2013</li> </ul> |
| PROFESSIONAL SERVICE | <ul style="list-style-type: none"> <li>• Beisitzer <ul style="list-style-type: none"> <li>– Introduction to symplectic geometry, University of Bonn, Summer 2025</li> <li>– Course on Locally analytic representations and geometric representation theory, University of Bonn, Summer 2026</li> </ul> </li> <li>• Master's thesis second reader, University of Bonn, Summer 2025</li> <li>• Topology Screening Exam Committee, University of Southern California, Fall 2024</li> <li>• Co-organizer, The Los Angeles Workshop on Representations and Geometry (LAWRGe), Summer 2023</li> <li>• Referee for Annals of Mathematics, Compositio, Geometry &amp; Topology, Selecta, and Journal of Topology (<math>\times 2</math>).</li> </ul>                                                                           |
| AWARDS AND HONORS    | <ul style="list-style-type: none"> <li>• Shing-Tung Yau College Student Mathematics Contest, Bronze medal in geometry and topology, 2014</li> <li>• Honors Program in Mathematics, National Taiwan University, 2014</li> <li>• Chow Hung-Ching Scholarship, Institute of Mathematics, Academia Sinica, 2013</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |